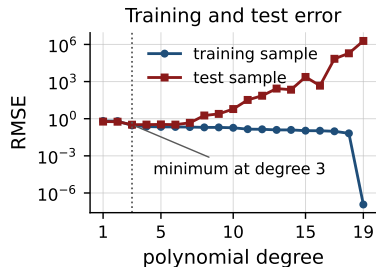
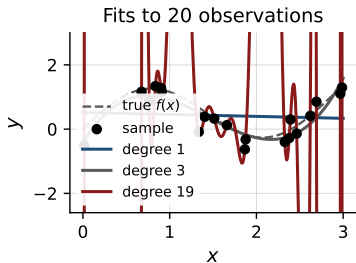


Prediction for Economists

Prediction as an objective

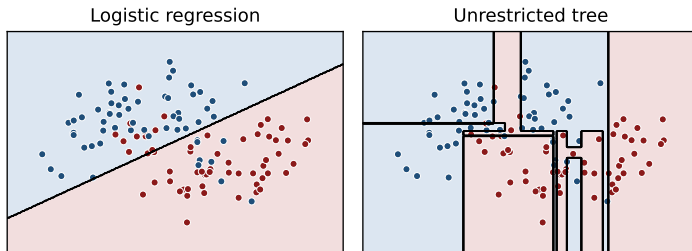
- A prediction rule maps observed characteristics into an unobserved outcome.
- It is estimated on a sample and judged on observations not used to estimate it.
- The criterion is accuracy out of sample, not the value of any coefficient.
- A model may describe the data poorly and still predict well.
- Bias in the coefficients is acceptable if it lowers prediction error.

Overfitting



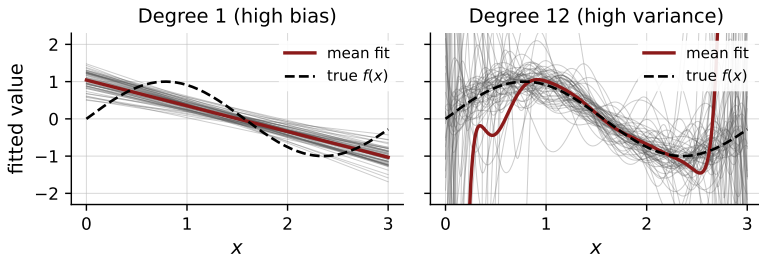
- Fit in the estimation sample improves monotonically with the degree.
- With 20 observations, degree 19 passes through every point exactly.
- Error in an independent sample is minimized at a low degree.

Overfitting with a binary outcome



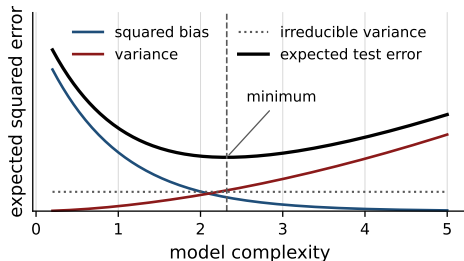
- The same arises when the outcome is a category, not a quantity.
- The unrestricted tree isolates individual observations.
- Its boundary reflects this sample, not a lasting regularity.

Bias and variance



- Each grey line is a fit to an independent sample of 25 observations.
- The restrictive fit varies little, but its mean is far from the truth.
- The flexible fit is right on average; any one estimate may be far off.

The bias–variance trade-off



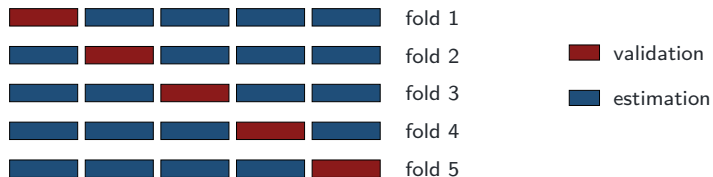
- Squared bias, variance, and noise.
- Restrictive fits: bias dominates.
- Flexible fits: variance dominates.
- The noise cannot be reduced.
- Choose a point on this curve.

Training, validation, and test samples



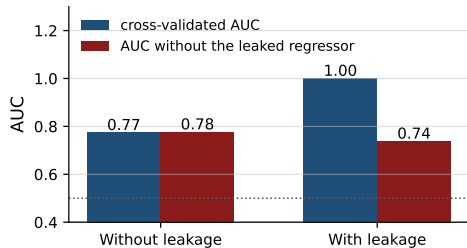
- Training: estimate the parameters of a given specification.
- Validation: choose among specifications, and how much flexibility.
- Test: used once, after every choice is final.
- Error in a sample used to choose is a downward-biased estimate.

Cross-validation



- Partition the sample into k subsets of equal size.
- Estimate on $k - 1$ subsets and evaluate on the one left out.
- Repeat so each subset serves once, then average the k estimates.
- With a binary outcome, folds preserve the proportion of ones.

Data leakage



- One regressor encodes the outcome.
- Its cross-validated fit is perfect.
- Without it, no better than honest.

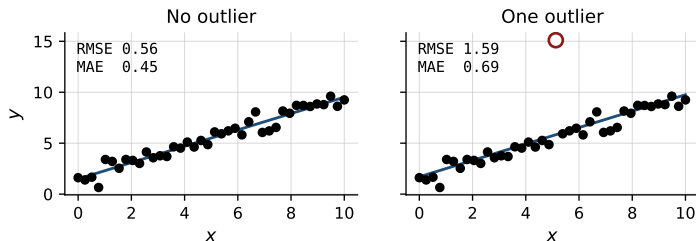
Where leakage enters

- Transformations computed on the full sample before it is split.
- Regressors dated after the outcome is realized.
- Any variable constructed from the outcome, including category means.
- A fit far better than expected is more often leakage than a good model.

Obtaining an honest estimate of performance

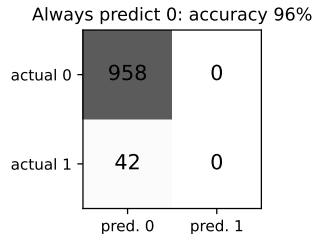
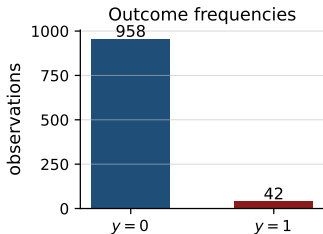
- Split the sample before computing any transformation.
- Estimate standardization and imputation on training observations only.
- Use the test sample once, after every choice has been made.
- Include a regressor only if it is known before the outcome is realized.

Measuring accuracy: continuous outcomes



- Root mean squared error weights large errors more heavily.
- Mean absolute error weights every unit of error equally.
- A few extreme observations move the first much more than the second.
- Fix the measure in advance, to reflect the cost of an error.

Measuring accuracy: binary outcomes

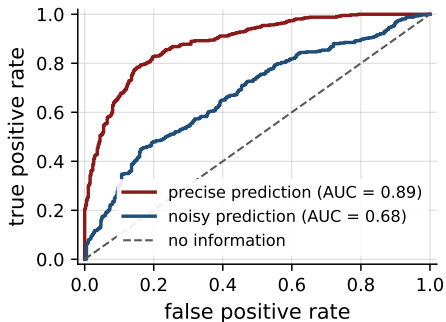


- When ones are rare, the share of correct classifications is uninformative.
- Predicting zero everywhere is 96% accurate and identifies nothing.
- The table of predicted against actual outcomes is the useful summary.

Thresholds, precision, and recall

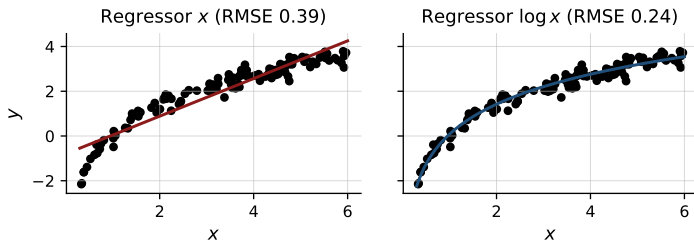
- Most estimators return a predicted probability, not a classification.
- Turning it into a classification requires a threshold.
- Precision: the share of predicted ones that are actual ones.
- Recall: the share of actual ones that are predicted.
- Raising the threshold raises precision and lowers recall.

The ROC curve



- Traces both rates over all thresholds.
- Its area summarizes the ordering.
- No single threshold is involved.
- A diagonal means no information.

The choice of regressors



- Both panels report least squares on a single regressor.
- The outcome is generated by $y = 2 \log x + \varepsilon$.
- Replacing x by $\log x$ lowers out-of-sample error by an order of magnitude.
- How the regressors are built often matters more than the estimator.

Constructing regressors in practice

- Indicators for categorical variables.
- Interactions and ratios.
- Logarithms and powers, for skewness and curvature.
- Discretization of continuous variables.
- Components of dates.
- An explicit indicator for missing values.

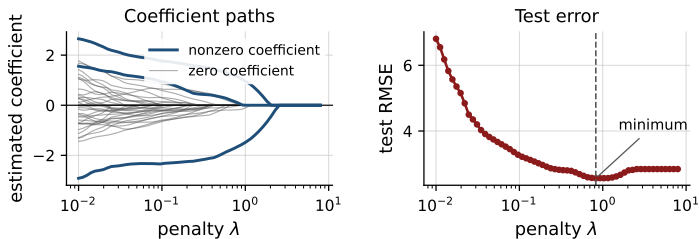
Two cautions

- Replacing a category by its mean outcome is a common source of leakage.
- It has to be computed separately within each training fold.
- Regressors with no predictive content raise the variance of the rule.
- Adding them is not free, even when their coefficients are near zero.

A menu of estimators

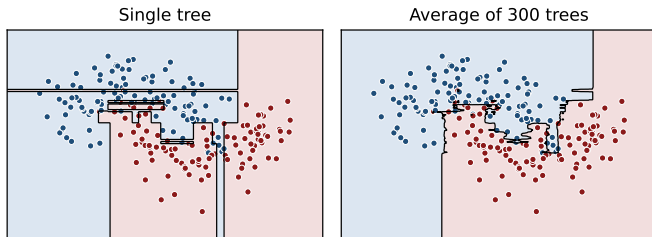
- Least squares: transparent, and often competitive with good regressors.
- Ridge and LASSO: a penalty on the coefficients; LASSO sets some to zero.
- Regression tree: a recursive partition; a single tree is unstable.
- Random forest: averages trees on resampled data; much lower variance.
- Boosting: small trees in sequence, each fitted to the residuals.
- Each has a counterpart when the outcome is binary.

Penalized regression



- 70 observations, 40 regressors, 3 with nonzero coefficients.
- The penalty shrinks coefficients and sets irrelevant ones to zero.
- Test error is minimized at a strictly positive penalty.
- The penalty is chosen by cross-validation, not by the fit.

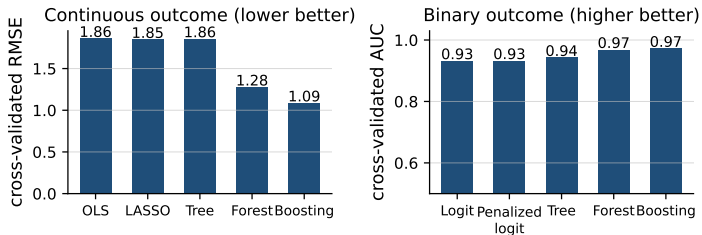
Trees and random forests



- A single tree's boundary depends heavily on the particular sample.
- A forest averages many trees, each estimated on resampled data.
- Flexibility is retained while the variance is substantially reduced.

- Trees are added in sequence rather than averaged in parallel.
- Each is fitted to the residuals of the current model.
- Each enters with a small weight, so progress is gradual.
- It is typically the most accurate of the estimators considered here.
- It has more parameters, and overfits if they are chosen carelessly.
- A random forest needs fewer choices and is a good starting point.

Comparing estimators



- The same regressors and the same folds for every estimator.
- The flexible estimators win here: the data have curvature and interaction.
- On data close to linear the ordering is frequently reversed.

The workflow



- Was every transformation estimated on training observations only?
- Could any regressor incorporate the outcome?
- Are the regressors and the folds identical across specifications?
- Prefer the simplest specification within sampling error of the best.

Summary

- Predictive accuracy concerns observations not used in estimation.
- Fit in the estimation sample does not establish it.
- The relevant trade-off is between bias and variance.
- Cross-validation is the standard way of estimating that accuracy.
- Fix the measure of accuracy in advance, by the cost of an error.
- The construction of regressors often matters more than the estimator.

Appendix

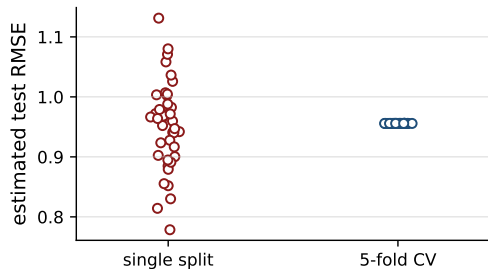
The decomposition of expected prediction error

For a new observation at x_0 , with $y = f(x_0) + \varepsilon$ and $\hat{f}$ estimated on a random sample,

$$\mathbb{E}[(y - \hat{f}(x_0))^2] = (\mathbb{E}[\hat{f}(x_0)] - f(x_0))^2 + \text{Var}(\hat{f}(x_0)) + \text{Var}(\varepsilon).$$

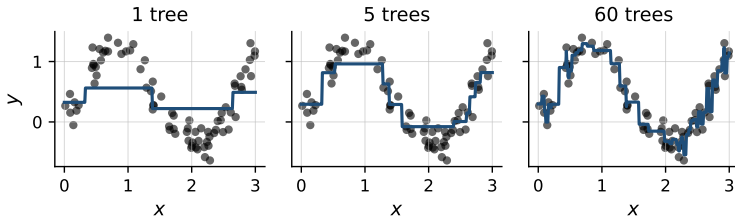
- The expectation is taken over samples, not over observations.
- The first two terms can be traded against one another.
- The third cannot be reduced by any estimator.
- Penalization and averaging lower the second at some cost in the first.

Sampling variability of a single split



- Each point uses a new partition.
- One split varies a great deal.
- The average over folds is stable.
- It matters most in small samples.

Boosting: successive fits



- A single tree of depth two is a coarse step function.
- Successive trees fitted to the residuals accumulate into an accurate fit.
- Continued far enough, the procedure begins to fit the noise.
- The number of trees is itself chosen by cross-validation.

Further measures for binary outcomes

- When ones are rare, the precision–recall curve is more informative.
- ROC can look favourable when few of the predicted ones are correct.
- The F_1 measure combines precision and recall in one number.
- Log-loss penalizes confident predictions that turn out to be wrong.
- Calibration: of those given probability 0.7, about 70% should be ones.

References

- Mullainathan and Spiess (2017), *Journal of Economic Perspectives* 31(2).
- Athey and Imbens (2019), *Annual Review of Economics* 11, 685–725.
- James, Witten, Hastie and Tibshirani, *Introduction to Statistical Learning*.
- Hastie, Tibshirani and Friedman, *The Elements of Statistical Learning*.